

# **Evolutionary Conservation of BRCA1 Across Primate Species Reveals High Constraint in Functional Domains**

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## Abstract

BRCA1 is a tumor suppressor gene essential for DNA repair, genomic stability, and cell cycle regulation. Evolutionary conservation across species can identify functionally important residues and structural domains. In this study, BRCA1 protein sequences from five primate species (*Homo sapiens*, *Pan troglodytes*, *Nomascus leucogenys*, *Macaca mulatta*, and *Papio anubis*) were compared using multiple sequence alignment, pairwise identity analysis, conservation scoring, and phylogenetic reconstruction. The RING and BRCT domains showed the highest conservation across all species, while the central region exhibited greater variability. Pairwise identity values ranged from 93–98 percent, decreasing with evolutionary distance from humans. Phylogenetic tree topology accurately reflected established primate evolution, supporting the reliability of BRCA1 as an evolutionary marker. These findings reinforce that evolutionary conservation strongly correlates with BRCA1's essential structural and enzymatic functions.

## Introduction

BRCA1 (Breast Cancer Type 1 Susceptibility Protein) is a multifunctional tumor suppressor involved in DNA double-strand break repair, checkpoint control, chromatin remodeling, and transcriptional regulation (Roy et al., 2012). Pathogenic variants in BRCA1 significantly increase the risk of hereditary breast and ovarian cancer, making it one of the most clinically relevant genes for genomic medicine (Foulkes, 2008). Given its critical role in maintaining genome

integrity, BRCA1 contains structurally defined domains, including the N-terminal RING domain and two C-terminal BRCT domains, that are essential for ubiquitin ligase activity and phospho-protein binding (Starita & Parvin, 2003; Williams et al., 2004).

Evolutionary conservation provides important insight into protein function. Amino acids that remain unchanged across long evolutionary periods are typically essential for stability or biochemical activity (Capra & Singh, 2007). High conservation in BRCA1 has been linked to functional constraint and to clinical interpretations of variant pathogenicity (Easton et al., 2007). Comparative genomics across primates offers a powerful approach to assess which regions of BRCA1 are under strong evolutionary pressure, because primate species share a recent evolutionary history but exhibit varying degrees of divergence (Perelman et al., 2011).

Previous studies have shown that while some regions of BRCA1 evolve rapidly, the core enzymatic and structural domains remain highly conserved (Huen et al., 2010). However, detailed conservation analyses across multiple primate species using full-length protein alignments remain limited. Understanding how BRCA1 diverges across primates can improve our knowledge of structure–function relationships and provide additional context for interpreting disease-associated mutations.

The aim of this project is to compare BRCA1 protein sequences across primates to identify conserved functional domains, quantify sequence identity, and assess evolutionary relationships using phylogenetic approaches.

## Objectives

1. To compare BRCA1 protein sequences across five primate species.
2. To perform multiple sequence alignment to identify conserved and variable amino acid regions.
3. To determine whether conserved regions correspond to known BRCA1 functional domains.
4. To construct a phylogenetic tree reflecting evolutionary relationships among primates.

## Methods

BRCA1 protein sequences were retrieved from NCBI RefSeq using canonical NP accessions for each species. The human reference sequence (NP\_009225.1) was used, and orthologs were collected for *Pan troglodytes*, *Nomascus leucogenys*, *Macaca mulatta*, and *Papio anubis*.

Predicted XP isoforms and alternate splice variants were excluded.

Multiple sequence alignment (MSA) was performed using COBALT with default parameters. The MSA was imported into Jalview for visualization, conservation scoring, and quality assessment. Conservation scores were calculated using Jalview's "Extract Scores → Conservation" tool, which computes residue conservation based on amino acid frequency.

Pairwise identity values were obtained using Jalview's global pairwise alignment function. A Neighbor-Joining phylogenetic tree (Poisson model, 100 bootstraps) was generated within Jalview from the aligned dataset. Figures were exported directly from Jalview.

## **Results/Analysis**

### **6.1 Multiple Sequence Alignment**

The alignment revealed high similarity across primate BRCA1 proteins, with the strongest conservation observed in the RING and BRCT domains. The RING domain (positions 24–64) showed nearly identical residues across all five species (Figure 1). In contrast, the central region of the protein displayed lower conservation, with multiple substitutions in macaque and baboon compared to humans and chimpanzees.

### **6.2 Pairwise Identity Matrix**

| Species 1 | Species 2 | % Identity    |
|-----------|-----------|---------------|
| Homo      | Pan       | <b>98.17%</b> |
| Homo      | Nomascus  | <b>96.03%</b> |
| Homo      | Macaca    | <b>93.03%</b> |
| Homo      | Papio     | <b>93.19%</b> |
| Pan       | Nomascus  | <b>96.73%</b> |
| Pan       | Macaca    | <b>93.45%</b> |
| Pan       | Papio     | <b>93.62%</b> |
| Nomascus  | Macaca    | <b>93.19%</b> |
| Nomascus  | Papio     | <b>93.29%</b> |
| Macaca    | Papio     | <b>98.87%</b> |

Identity decreased with evolutionary distance, consistent with primate evolutionary history.

### 6.3 Conservation Analysis

The RING domain exhibited uniformly high conservation and strong consensus agreement (Figure 1). The central region showed lower conservation and greater variability (Figure 2). The C-terminal BRCT1 and BRCT2 domains (positions ~1650–1850) displayed pronounced conservation peaks, indicating strong functional constraint (Figure 3).

### 6.4 Phylogenetic Tree

The phylogenetic tree accurately separated apes from Old World monkeys, grouping humans with chimpanzees and placing gibbon as an intermediate branch. Macaque and baboon clustered together, consistent with established primate phylogeny.

## **Discussion**

The results demonstrate that BRCA1 is highly conserved across primates, particularly within its functional domains. The RING domain exhibited near-complete conservation, consistent with its role in ubiquitin ligase activity and protein–protein interactions essential for DNA repair (Brzovic et al., 2003). Such strong conservation suggests that even minor amino acid changes in this region could disrupt BRCA1 function, aligning with clinical observations that many pathogenic variants cluster within or near the RING domain (Starita et al., 2015).

The central region displayed lower conservation, supporting previous findings that this segment of BRCA1 lacks stable tertiary structure and is more tolerant to evolutionary variation (Mallery et al., 2002). Reduced constraint in this region suggests a lesser impact of substitutions on BRCA1's structural integrity.

The BRCT domains showed high conservation across all species, reflecting their critical role in phospho-protein binding during DNA damage response (Williams et al., 2004). This high conservation aligns with extensive biochemical evidence showing that BRCT domain mutations frequently cause loss of BRCA1 function and predispose individuals to hereditary cancers (Easton et al., 2007).

Finally, the phylogenetic tree recovered expected primate relationships, reinforcing that BRCA1 evolves under strong purifying selection and retains phylogenetic signal despite its essential function. These findings collectively support the hypothesis that evolutionary conservation strongly correlates with functional importance in BRCA1.

## **Conclusion**

Comparative analysis of BRCA1 across five primate species revealed strong evolutionary conservation in structurally and functionally important domains, including the RING and BRCT repeats. Pairwise identity values and phylogenetic clustering were consistent with established primate divergence patterns. The high conservation of BRCA1's functional domains highlights their essential role in DNA repair and genomic integrity. These results provide evolutionary context that may aid interpretation of BRCA1 variants in clinical genetics.



## References

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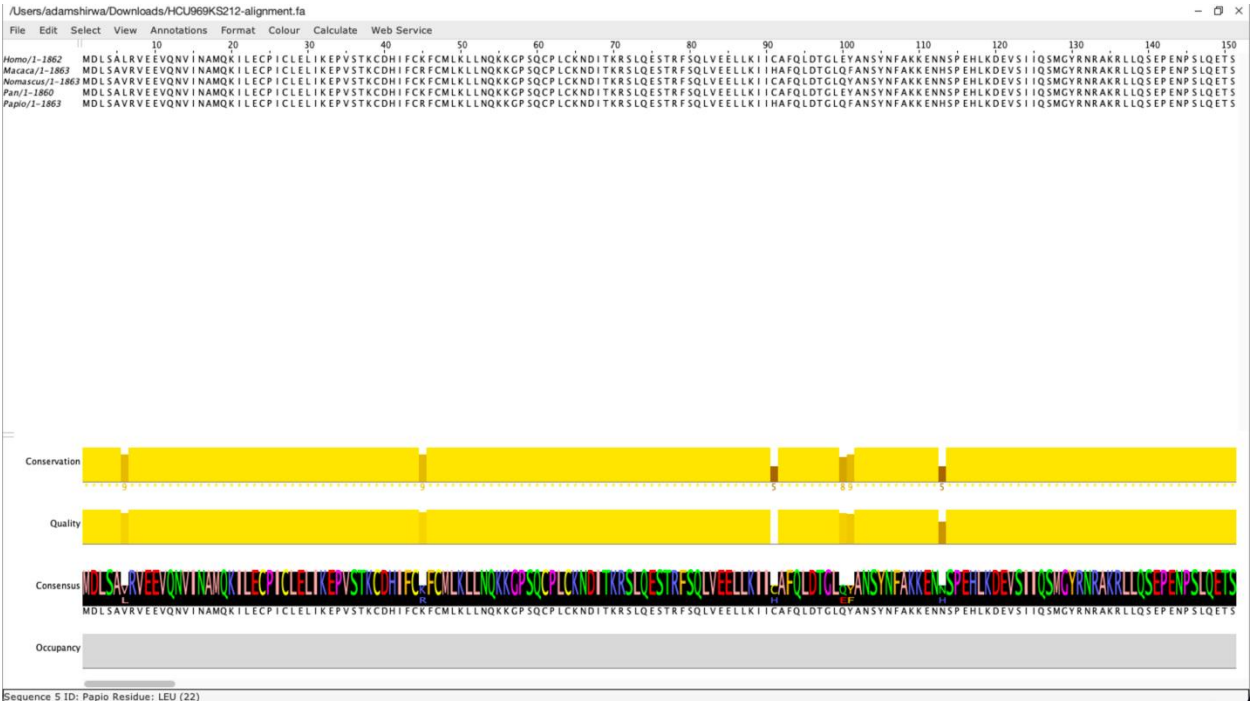
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Appendix

Appendix A. Full Multiple Sequence Alignment



Appendix B: Full FASTA Sequence Text

>Homo sapiens

MDLSALRVEEVQNVINAMQKILECPICLELIKEPVSTKCDHIFCKFCMLKLLNQKKGPSQ  
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HSHY

>Macaca mulatta

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>Nomascus leucogenys

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>Pan paniscus

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>Papio anubis

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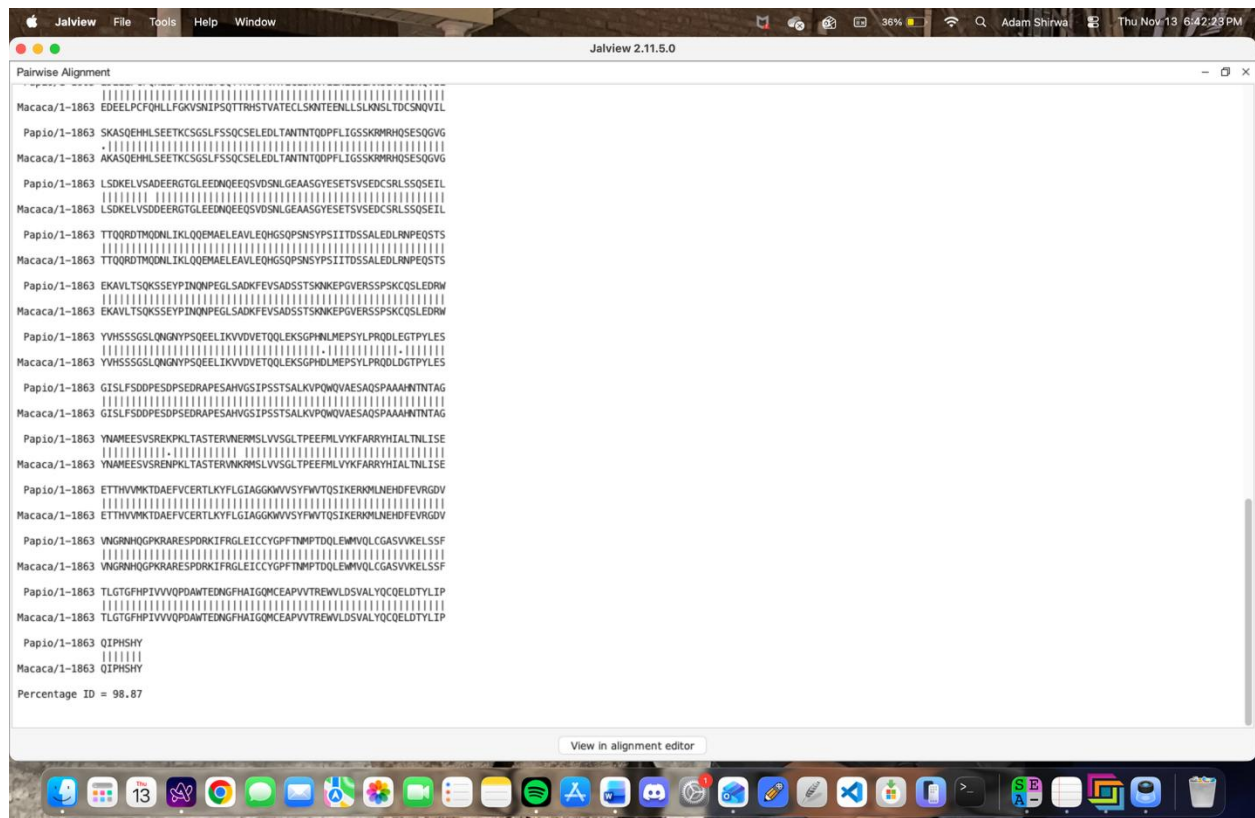


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TNTAGYNAM EESVSREKPKLTASTERNERMSLVVSGLTPEEFMLVYKFARRYHIALTNL  
ISEETTHVVMKTD AEFVCERTLK YFLGIAGGKWVVS YFWVTQSIKERKMLNEHDFEVRGD  
VVNGRNHQGPKRARESPDRKIFRGLEICCYGPFTNMPTDQLEWMVQLCGASVVKELSSFT  
LGTGFHPIVVVQPD AWTEDNGFHAIGQMCEAPVVTREWWLDSVALYQCQELDTYLIPQIP  
HSHY

# Appendix C. Full Conservation Profile



# Appendix D. Example Pairwise Global Alignment Windows



(Screenshot of Papio vs Macaca pairwise alignment window)

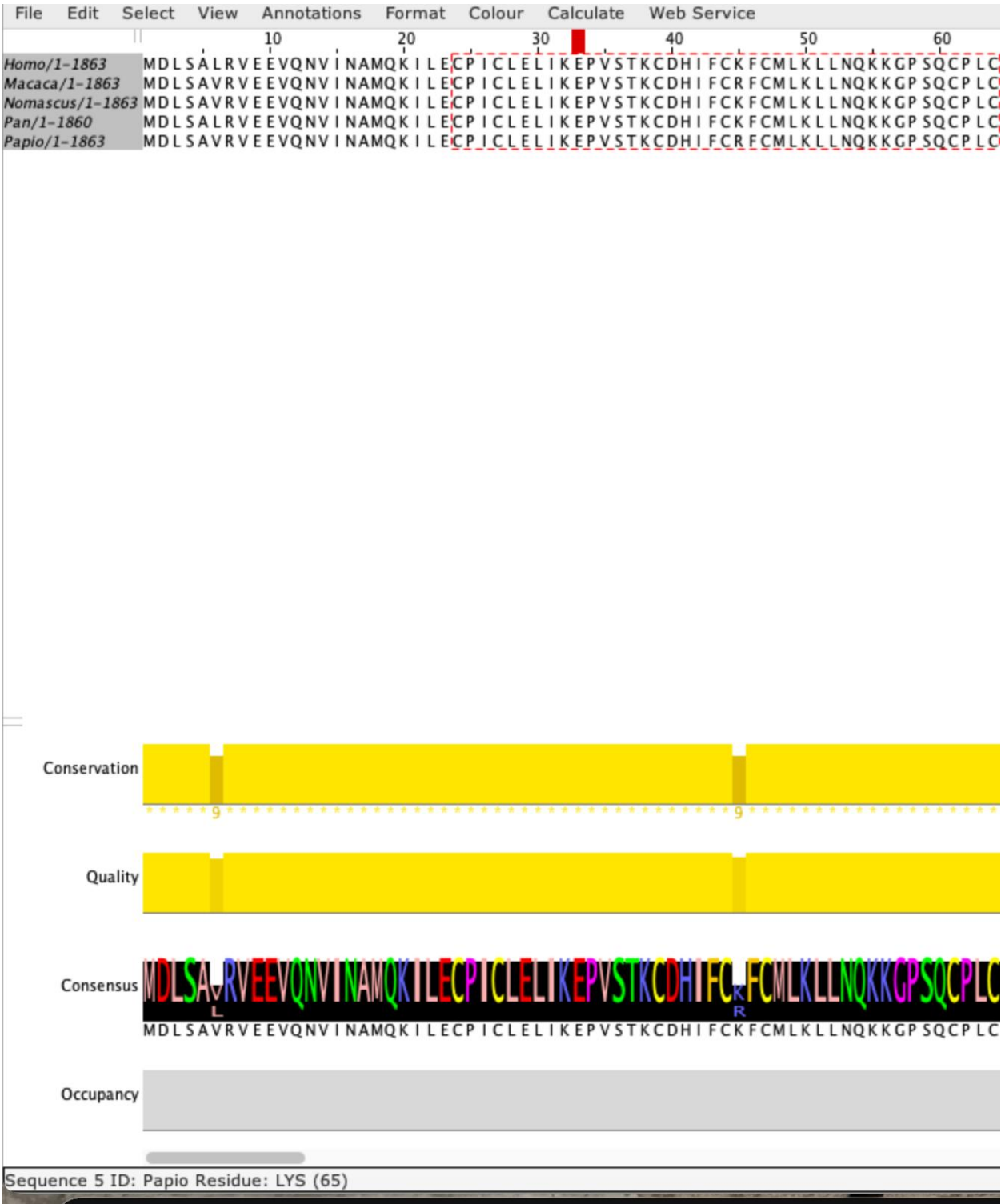
## Appendix E. Accession Numbers Used

The following RefSeq accessions were used to obtain BRCA1 orthologs:

- Homo sapiens NP\_009225.1
- Pan troglodytes NP\_001038958.1
- Nomascus leucogenys NP\_001295597.1
- Macaca mulatta NP\_001108421.1
- Papio anubis NP\_001289032.1

## Appendix F. Conservation in Key Functional Domains

B1. RING Domain (positions ~24–64)



Conservation



Quality



Consensus

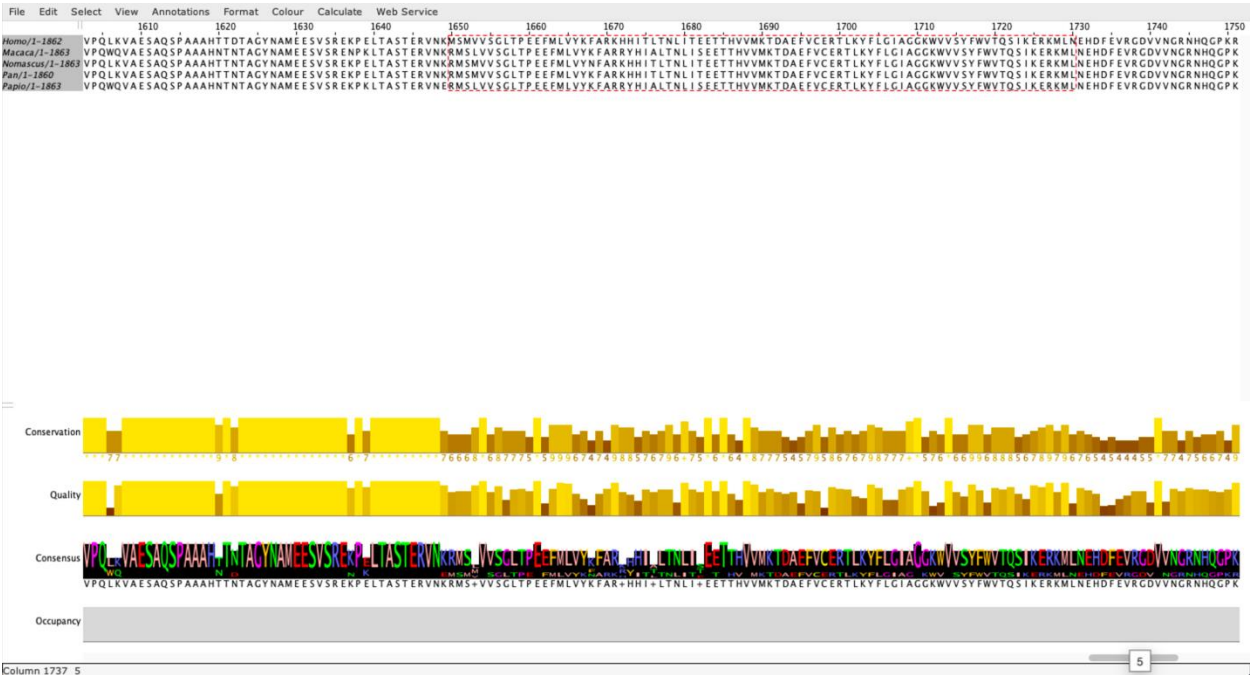


Occupancy



Sequence 5 ID: Papio Residue: LYS (65)

B2. BRCT1 Domain (positions ~1650–1730)



B3. BRCT2 Domain (positions ~1760–1850)

